1. The head loss in turbulent flow in a pipe varies
   - A. as velocity
   - B. as $(\text{velocity})^2$
   - C. inversely as the square of diameter
   - D. inversely as the velocity

[#]

2. The drag co-efficient for a bacterium moving in water at 1 mm/s, will be of the following order of magnitude (assume size of the bacterium to be 1 micron and kinematic viscosity of water to be $10^{-6} m^2/s$).
   - A. 24000
   - B. 24
   - C. 0.24
   - D. 0.44

[#]

3. For the free settling of a spherical particle through a fluid, the slope of, C_D-log N_{Re} , plot is
   - A. 1
   - B. -1
   - C. 0.5
   - D. -0.5

[#]

4. Hydraulic intensifier is used for increasing the
   - A. rate of velocity of liquid supply.
   - B. rate of flow through delivery pipeline of a pump.
   - C. intensity of pressure of the liquid.
   - D. momentum rate through delivery pipe.

[#]

5. Transition length for a turbulent fluid entering into a pipe is around _____ times the pipe diameter.
   - A. 5
   - B. 50
   - C. 500
   - D. 1000

[#]

6. Pascal's law is valid, only when the fluid is

- [ ] A. frictionless and at rest.
- [ ] B. at rest.
- [ ] C. at rest and when the frictionless fluid is in motion.
- [ ] D. none of these.

[#]

7. Pick out the Kozney-Karman equation (valid for low N_{Re}) for fluid flow through a packed bed of solids.

- [ ] A. $\frac{\Delta p}{\rho} = 4f \cdot \frac{L}{D} \cdot \frac{V^2}{2gc}$
- [ ] B. $f_p = \frac{150(1-\varepsilon)}{N_{Re}} + 1.75$
- [ ] C. $\frac{-\Delta p \cdot gc \cdot D^2_p \cdot \varepsilon^3}{L \cdot \overline{V}_0 \cdot \mu (1-\varepsilon)^2} = 150$
- [ ] D. $\frac{-\Delta p}{\delta \cdot L} \cdot \frac{gc}{V^2_0} \cdot \frac{D_p \cdot \varepsilon^3}{1-\varepsilon} = 1.75$

[#]

8. One dimensional fluid flow implies the

- [ ] A. flow in straight lines only.
- [ ] B. uniform flow.
- [ ] C. steady uniform flow.
- [ ] D. flow in which transverse components are zero.

[#]

9. The discharge through a venturimeter depends upon

- [ ] A. pressure drop only.
- [ ] B. its orientation.
- [ ] C. co-efficient of contraction only.
- [ ] D. none of these.

[#]

10. The pressure and power requirement of a gas fan at constant speed & capacity varies _____ the gas density.

- [ ] A. directly as

- B. inversely as square root of
- C. inversely as
- D. as square of

[#]

11. Gradually varied flow in open channel is a/an _____ flow.
    - A. steady uniform
    - B. steady non-uniform
    - C. unsteady uniform
    - D. unsteady non-uniform

[#]

12. At low Reynolds number
    - A. viscous forces are unimportant.
    - B. viscous forces control.
    - C. viscous forces control and inertial forces are unimportant.
    - D. gravity forces control.

[#]

13. Buckingham-π theorem states that in any physical problem including '*n*' quantities having '*m*' diamensions, the quantities can be arranged into _____ independent dimensionless parameters.
    - A. *m*
    - B. *n*
    - C. *n*-*m*
    - D. *n*/*m*

[#]

14. Pressure difference between two points in vessels, pipelines or in two different pipelines can be measured by a differential manometer. The pressure difference measured as the mm of water coloumn in case of mercury-wa

- B. deep
- C. both (a) & (b)
- D. neither (a) nor (b)

[#]

20. The Reynolds number for an ideal fluid flow is

- A. 4
- B. 2100-4000
- C. 4000
- D. ∞

[#]